

Forms of Maxwell's Equation (General Expression or Dynamic or Time-Varying)

Integral Form

Point form

$$\oint_S \vec{E} \circ \hat{n} da = q_{\text{in}}/\epsilon_0 \xrightarrow[\text{theorem}]{\text{Divergence}} \vec{\nabla} \circ \vec{E} = \rho/\epsilon_0 \quad \text{Gauss's law for electric fields}$$

$$\oint_S \vec{B} \circ \hat{n} da = 0 \xrightarrow[\text{theorem}]{\text{Divergence}} \vec{\nabla} \circ \vec{B} = 0 \quad \text{Gauss's law for magnetic fields}$$

$$\oint_C \vec{E} \circ d\vec{l} = -\frac{d}{dt} \int_S \vec{B} \circ \hat{n} da \xrightarrow[\text{theorem}]{\text{Stokes'}} \vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad \text{Faraday's law}$$

$$\oint_C \vec{B} \circ d\vec{l} = \mu_0 \left(I_{\text{enc}} + \epsilon_0 \frac{d}{dt} \int_S \vec{E} \circ \hat{n} da \right) \xrightarrow[\text{theorem}]{\text{Stokes'}} \vec{\nabla} \times \vec{B} = \mu_0 \left(\vec{J} + \epsilon_0 \frac{\partial \vec{E}}{\partial t} \right) \quad \text{Ampere-Maxwell law}$$

*in a source-free region $\rho = 0$ and $\vec{J} = 0$

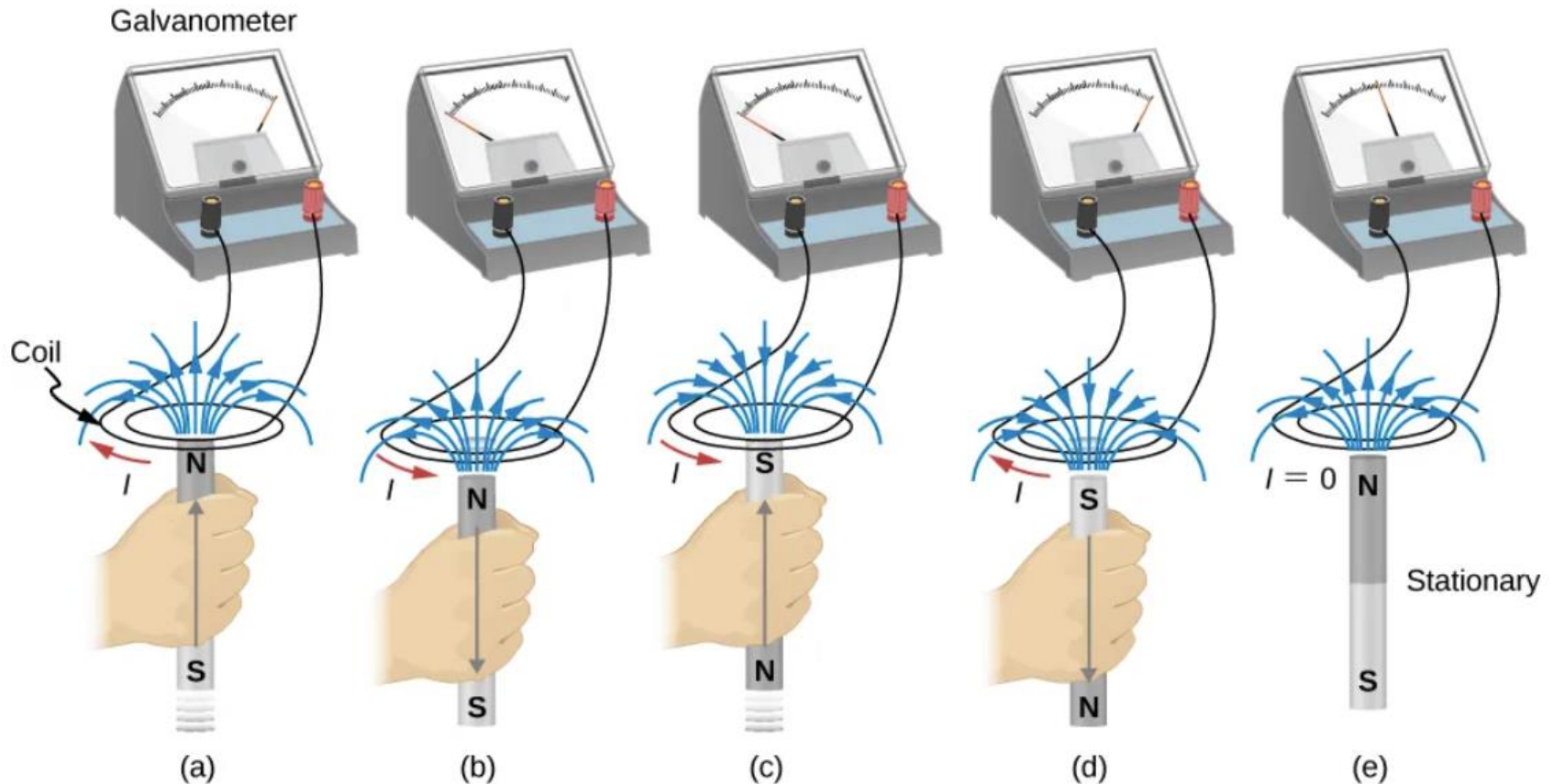
**for static $\frac{d}{dt} = 0$

Recap

- In Maxwell's Equations, you'll encounter two kinds of electric field: the **electrostatic field** produced by electric charge and the **induced electric field** produced by a changing magnetic field.
- Similarly, there are two fundamental types of magnetic fields: **Static (or Permanent) Magnetic Field**: This type of magnetic field is associated with permanent magnets or steady currents and **Induced (or Time-Varying) Magnetic Field**: This type of magnetic field is produced by changing electric fields, according to the amended Ampère's law, which incorporates the concept of displacement current.

Faraday's Law (1831) – Observations

A TIME VARYING MAGNETIC FLUX
PRODUCES AN ELECTRIC CURRENT.

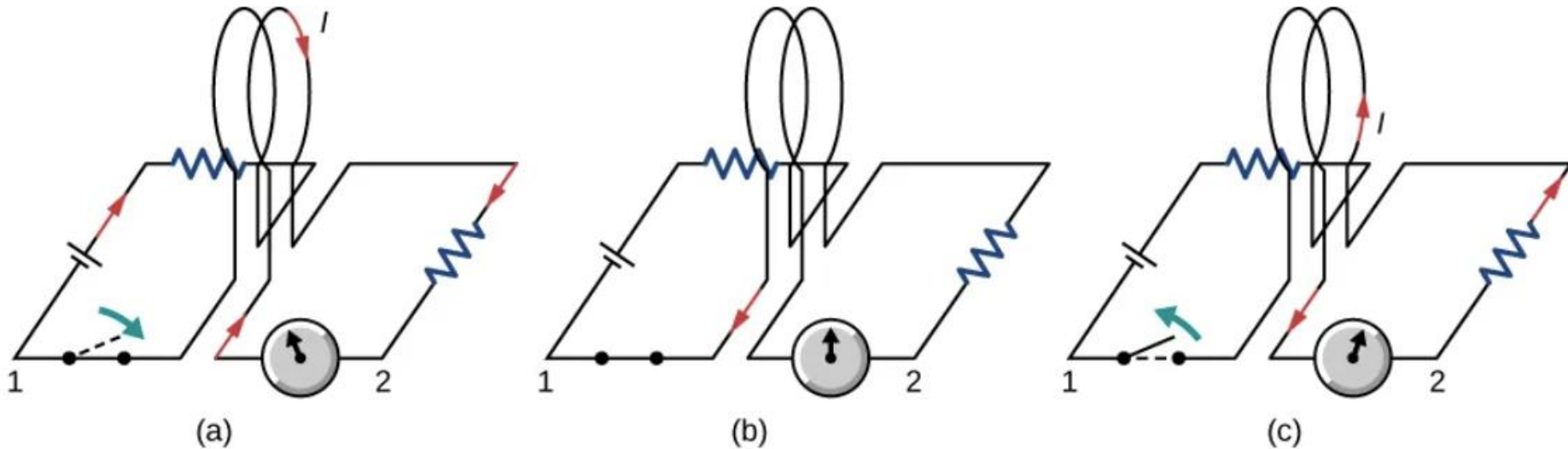


Movement of a magnet relative to a coil produces emfs as shown (a–d). The same emfs are produced if the coil is moved relative to the magnet. This short-lived emf is only present during the motion. The greater the speed, the greater the magnitude of the emf, and the emf is zero when there is no motion, as shown in (e).

Faraday's Law (1831) – Observations

A TIME VARYING MAGNETIC FLUX
PRODUCES AN ELECTRIC CURRENT.

Faraday also discovered that a similar effect can be produced using two circuits—a changing current in one circuit induces a current in a second, nearby circuit.



(a) Closing the switch of circuit 1 produces a short-lived current surge in circuit 2.

(b) If the switch remains closed, no current is observed in circuit 2.

(c) Opening the switch again produces a short-lived current in circuit 2 but in the opposite direction from before.

Faraday's Law (1831)

- The magnetic flux linking the loop

$$\Phi_B = \int_S \vec{B} \cdot \hat{n} da.$$

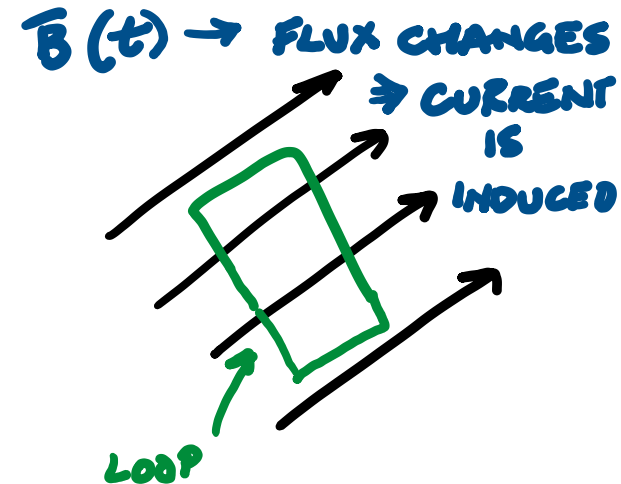
FARADAY'S LAW : $\text{emf} : \text{ELECTROMOTIVE FORCE}$

INDUCED VOLTAGE IN ANY CLOSED CIRCUIT IS EQUAL TO THE TIME RATE OF CHANGE OF MAGNETIC FLUX LINKAGE BY THE CIRCUIT $\rightarrow N \Phi_B$

$$V_{\text{emf}} = -N \frac{d\Phi_B}{dt}$$

Q: WHY -ve SIGN ?

N : NUMBER OF LOOPS ; Φ_B : FLUX



Faraday's Law (1831)

Electromotive force (voltage) induced by time-varying magnetic flux:

$$V_{\text{emf}} = -N \frac{d\Phi}{dt} = -N \frac{d}{dt} \int_S \mathbf{B} \cdot d\mathbf{s} \quad (\text{V})$$

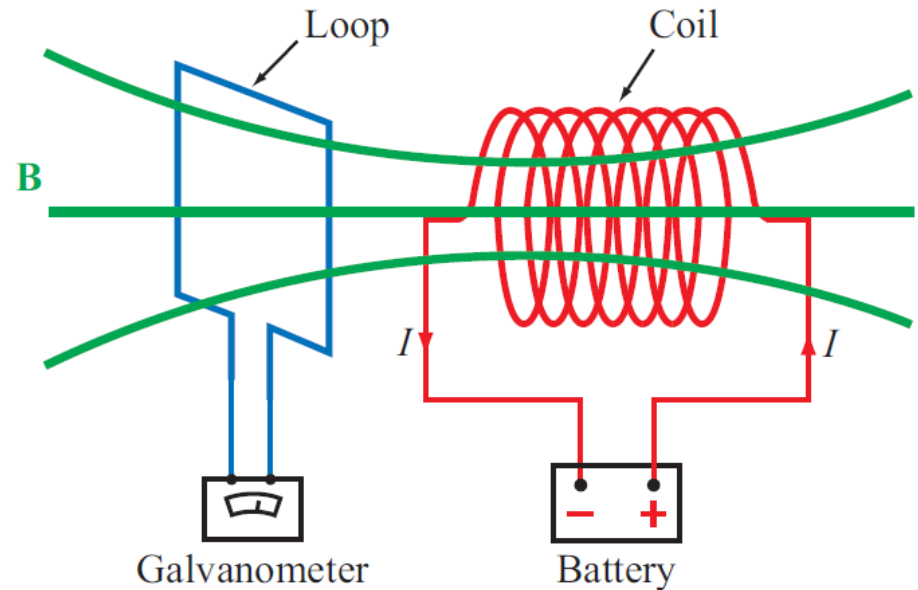


Figure 6-1: The galvanometer (predecessor of the ammeter) shows a deflection whenever the magnetic flux passing through the square loop changes with time.

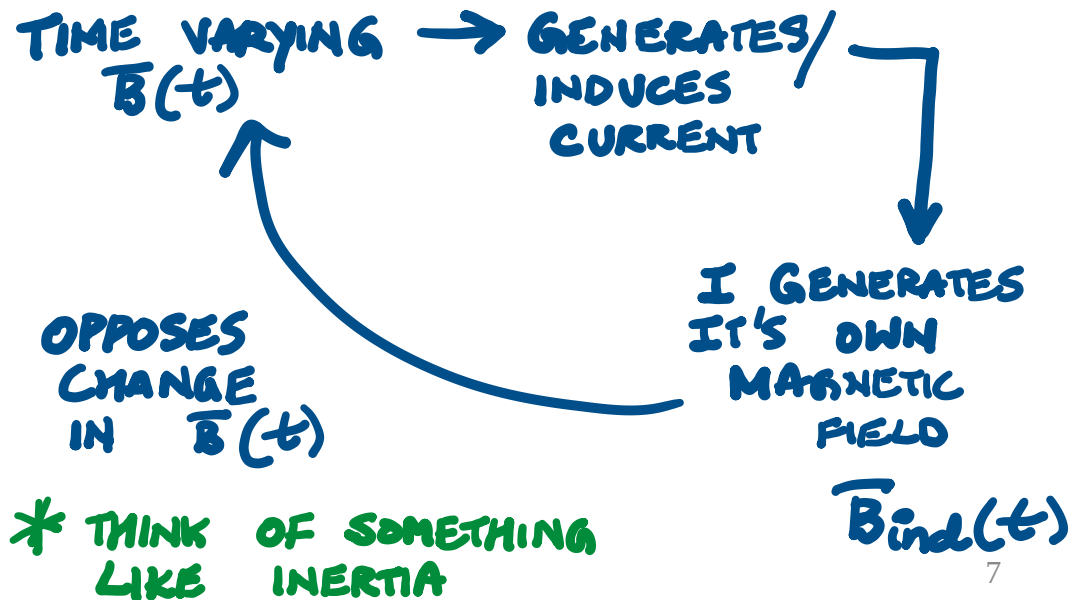
Magnetic fields can produce an electric current in a closed loop, but only if the magnetic flux linking the surface area of the loop changes with time. The key to the induction process is change.

WHY DOES FARADAY'S LAW HAVE A NEGATIVE SIGN?

THE INDUCED EMF ACTS TO OPPOSE THE FLUX PRODUCING IT

⇒ LENZ'S LAW

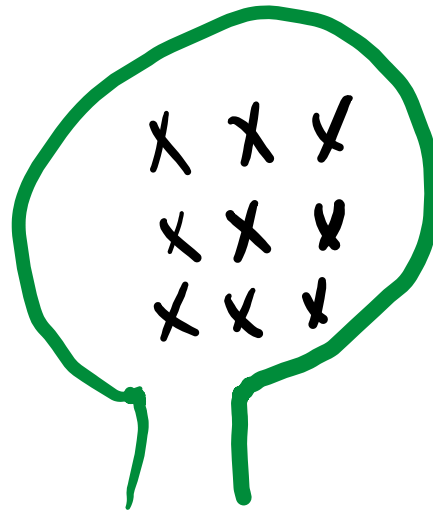
"THE DIRECTION OF CURRENT IN THE CIRCUIT IS SUCH THAT THE INDUCED MAGNETIC FIELD OPPOSES CHANGE IN THE ORIGINAL MAGNETIC FIELD"



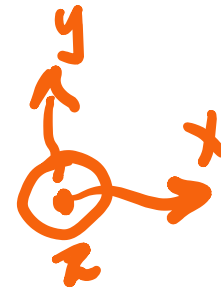
Example: Consider a coil of wire in a changing (with time) magnetic field that is uniform over the cross section of the wire loop. Determine the direction of the current in the loop in the two cases below.
 B is in the $-\hat{z}$ direction

a) $dB/dt > 0$

b) $dB/dt < 0$



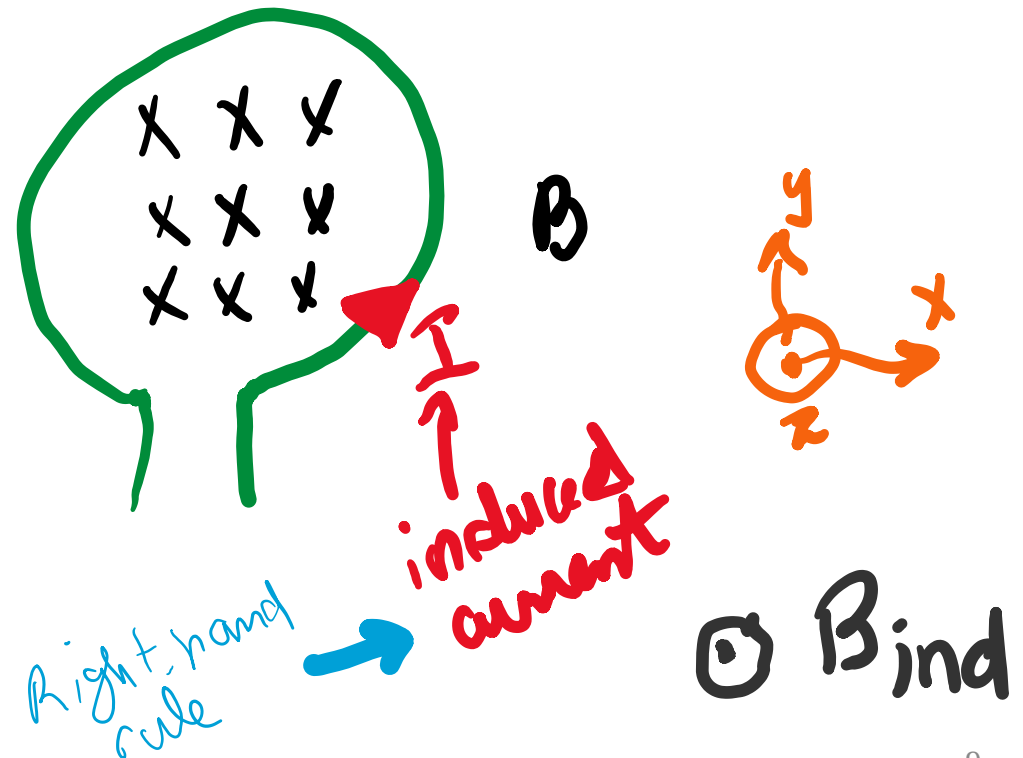
B



Example: Consider a coil of wire in a changing (with time) magnetic field that is uniform over the cross section of the wire loop. Determine the direction of the current in the loop in the two cases below.
 B is in the $-\hat{z}$ direction

a) $\frac{dB}{dt} > 0$

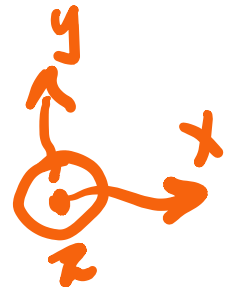
If $\frac{dB}{dt} > 0$, the flux in the coil increases.
The induced current in the coil will be counter-clockwise to **oppose the change** in the magnetic field



Example: Consider a coil of wire in a changing (with time) magnetic field that is uniform over the cross section of the wire loop. Determine the direction of the current in the loop in the two cases below.
 B is in the $-\hat{z}$ direction

b) $\frac{dB}{dt} < 0$

If $\frac{dB}{dt} < 0$, the flux in the coil decreases. The induced current in the coil will be clockwise to add more magnetic field in the $-\hat{z}$ direction and hence **oppose the change** in the magnetic field

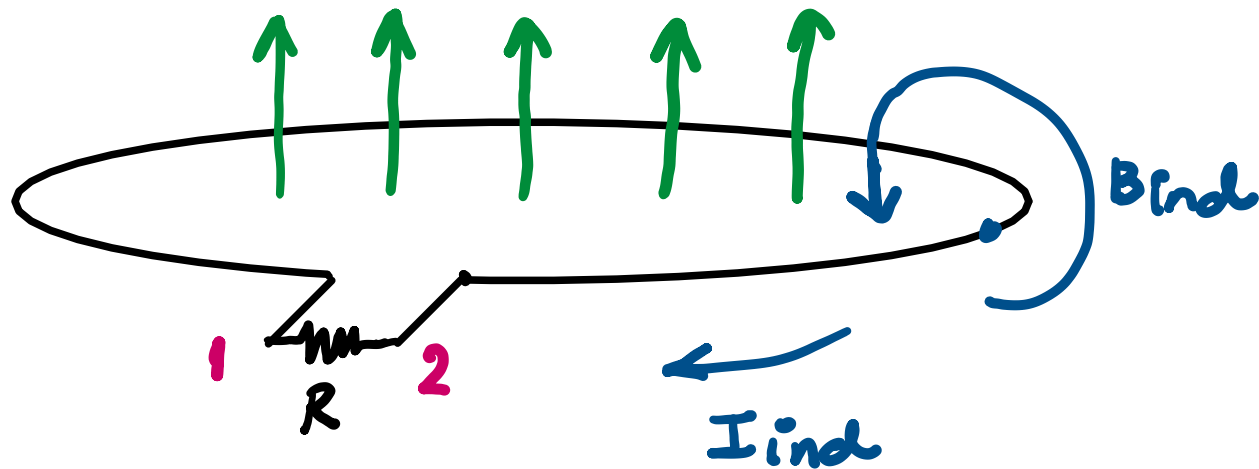


@ bind

EXAMPLE : LOOP PLACED IN
A CHANGING $\vec{B}$.

Q. WHAT IS THE DIRECTION OF
INDUCED CURRENT ?

$\vec{B}(t)$ INCREASES WITH t



RIGHT-HAND
RULE.

WHAT IS NEEDED TO INDUCE V_{emf} ?

$$V_{emf} = -N \frac{d\Phi}{dt} = -N \frac{d}{dt} \int_S \mathbf{B} \cdot d\mathbf{s} \quad (V)$$

SO TO INDUCE emf :

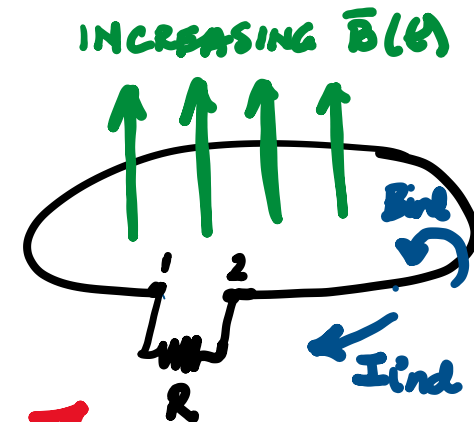
- (a) $d\bar{S}$ IS CONSTANT AND $\bar{B}$ VARIES
WITH t (TRANSFORMER emf)
- (b) $\bar{B}$ IS CONSTANT ; $d\bar{S}$ CHANGES
WITH TIME (MOTIONAL emf)
- (c) BOTH $\bar{B}$ AND $d\bar{S}$ CHANGE
WITH TIME

KEY THING IS CHANGE ! $V_{emf} = V_{emf}^{tr} + V_{emf}^m$

Transformer emf

STATIONARY $\vec{B}$ FIELD LOOP IN TIME VARYING (TRANSFORMER emf)

$$\Rightarrow \text{Transformer emf} = -N \iint_S \frac{\partial \vec{B}}{\partial t} \cdot d\vec{S}$$



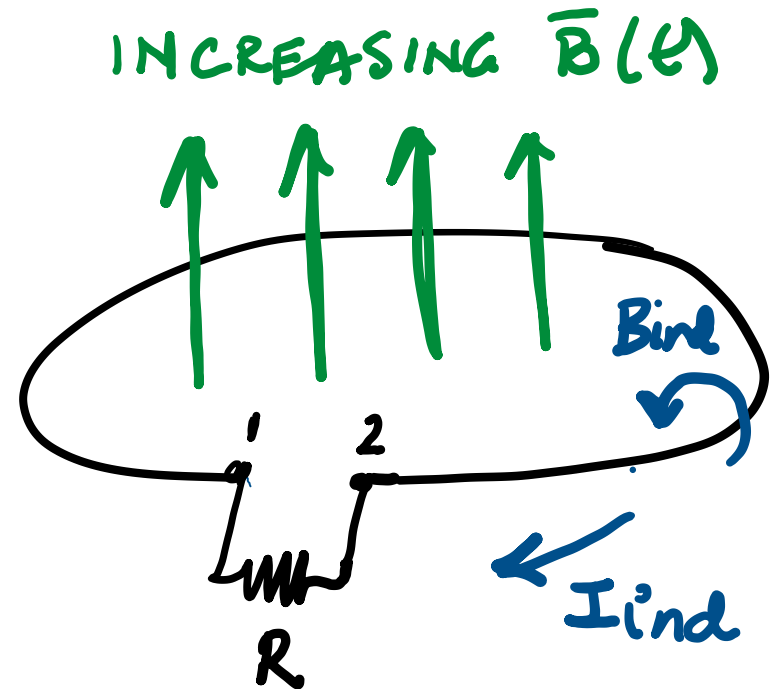
EXAMPLE OF
TRANSFORMER
emf.

Transformer emf

WHAT IS THE POLARITY OF V_{emf} ACROSS THE RESISTOR?

CONVENTION!

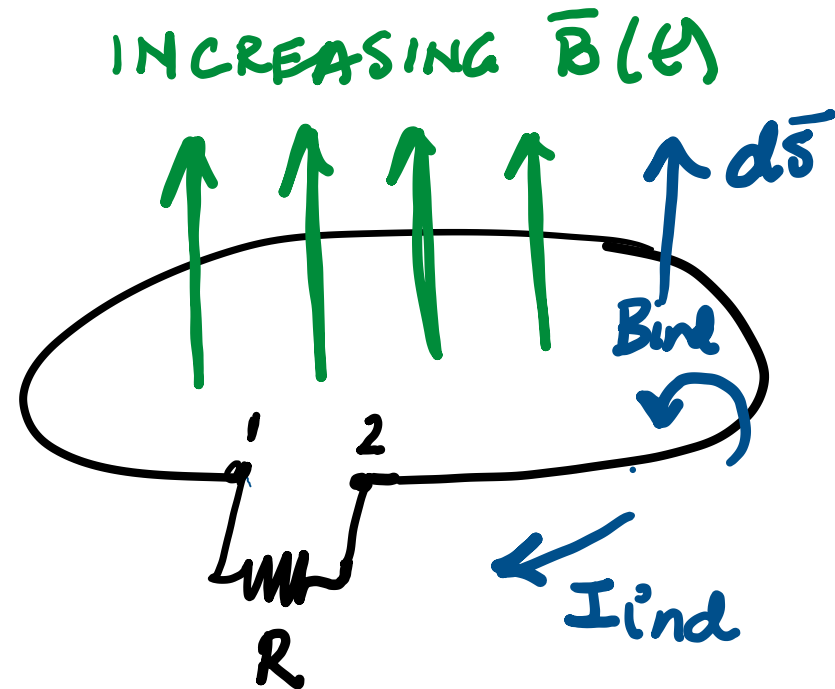
* THE POLARITY OF V_{emf} IS DETERMINED BY HOW WE CHOOSE IS NOT THE DIRECTION OF I



Transformer emf

WHAT IS THE POLARITY OF V_{cmf} ACROSS THE RESISTOR?

SAY $d\vec{s}$ IS POINTING UPWARDS

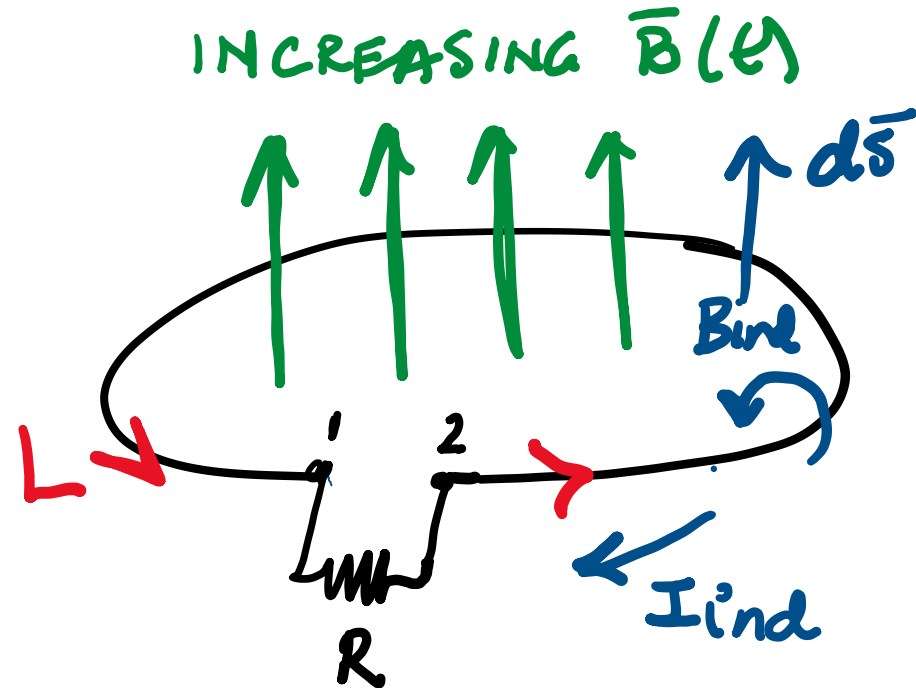


→ USE RIGHT AND RULE AND PLACE THUMB ALONG $d\vec{s}$

Transformer emf

WHAT IS THE POLARITY OF V_{emf} ACROSS THE RESISTOR?

SAY $d\vec{s}$ IS POINTING UPWARDS

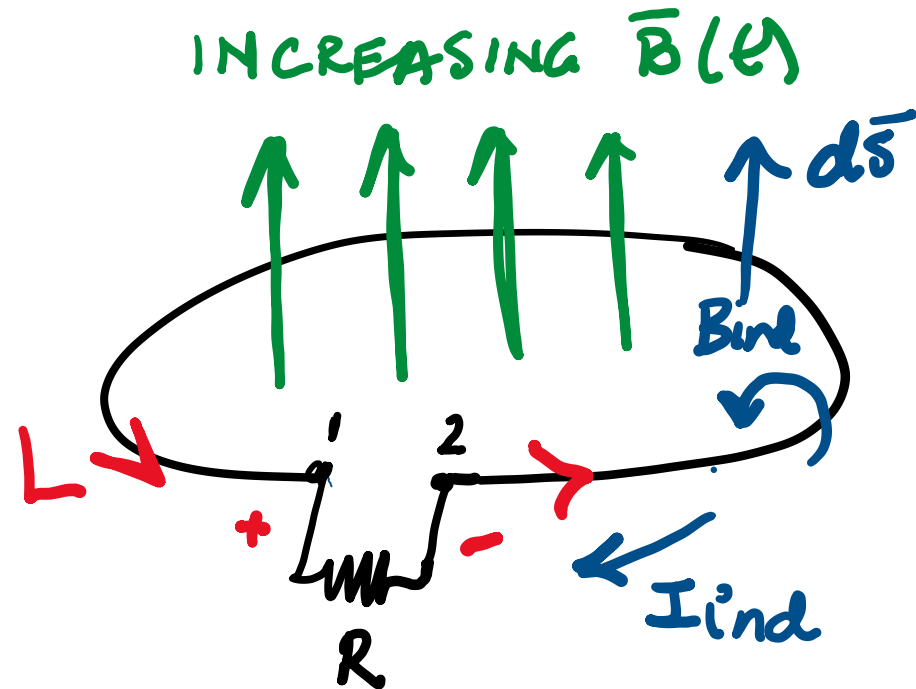


→ DIRECTION OF FINGERS GIVES US "DIRECTION OF LOOP" (L)

Transformer emf

WHAT IS THE POLARITY OF V_{emf} ACROSS THE RESISTOR?

SAY $d\vec{s}$ IS POINTING UPWARDS



→ TRAVEL ALONG L AND FIRST TERMINAL IT ENCOUNTERS IS $+$ ve.

⇒ TERMINAL 1 IS ASSUMED TO BE POSITIVE (AS PER CONVENTION).

Transformer emf

$$V_{\text{emf}}^{\text{tr}} = -N \int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s} \quad (\text{transformer emf}),$$

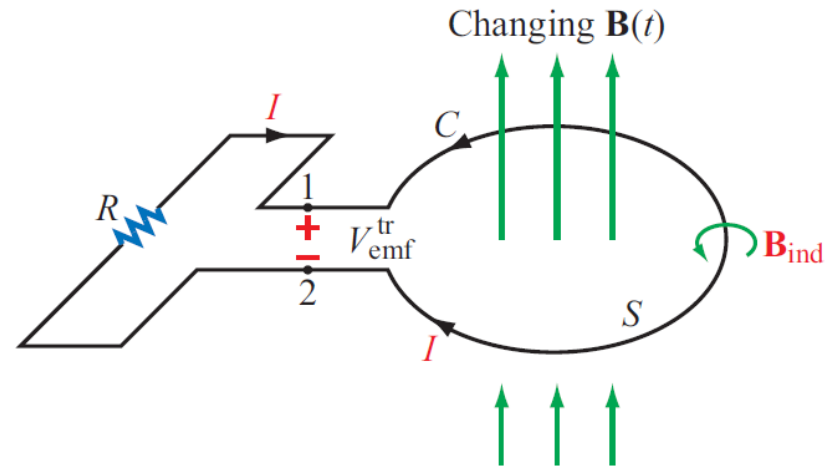
It is important to remember that $\mathbf{B}_{\text{ind}}$ serves to oppose the change in $\mathbf{B}(t)$, and not necessarily $\mathbf{B}(t)$ itself.

The connection between the direction of $d\mathbf{s}$ and the polarity of $V_{\text{emf}}^{\text{tr}}$ is governed by the following right-hand rule: if $d\mathbf{s}$ points along the thumb of the right hand, then the direction of the contour C indicated by the four fingers is such that it always passes across the opening from the positive terminal of $V_{\text{emf}}^{\text{tr}}$ to the negative terminal.

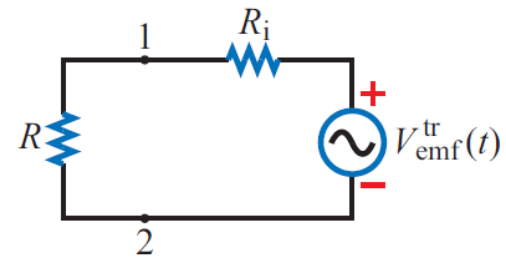
$$I = \frac{V_{\text{emf}}^{\text{tr}}}{R + R_i} . \quad (6.9)$$

For good conductors, R_i usually is very small, and it may be ignored in comparison with practical values of R .

*The polarity of $V_{\text{emf}}^{\text{tr}}$ and hence the direction of I is governed by **Lenz's law**, which states that the current in the loop is always in a direction that opposes the change of magnetic flux $\Phi(t)$ that produced I .*



(a) Loop in a changing $\mathbf{B}$ field



(b) Equivalent circuit

Figure 6-2: (a) Stationary circular loop in a changing magnetic field $\mathbf{B}(t)$, and (b) its equivalent circuit.

Faraday's Law as a Maxwell's Equation

TRANSFORMER emf ($V_{\text{emf}}^{\text{tr}}$)

$$V_{\text{emf}}^{\text{tr}} = - \iint_S \frac{\partial \bar{B}}{\partial t} \cdot d\bar{S} \quad [N=1]$$

ALSO WE CAN DEFINE V_{emf} IN TERMS OF

$\bar{E}$ AS

$$V_{\text{emf}} = \oint \bar{E} \cdot d\bar{\ell}$$

$\bar{E}$ PRODUCED BY V_{emf} IN THE WIRE

$$\Rightarrow \oint \underline{\bar{E} \cdot d\bar{\ell}} = - \iint_S \frac{\partial \bar{B}}{\partial t} \cdot d\bar{S}$$

Faraday's Law as a Maxwell's Equation

$$\Rightarrow \oint \underline{\underline{\vec{E} \cdot d\vec{l}}} = - \iint_S \underline{\underline{\frac{\partial \vec{B}}{\partial t} \cdot d\vec{S}}}$$

APPLYING STOKES THEOREM

$$\Rightarrow \iint_S \underline{\underline{(\vec{\nabla} \times \vec{E}) \cdot d\vec{S}}} = - \iint_S \underline{\underline{\frac{\partial \vec{B}}{\partial t} \cdot d\vec{S}}}$$

$$\boxed{\vec{\nabla} \times \vec{E} = - \frac{\partial \vec{B}}{\partial t}}$$

* DIFFERENTIAL FORM OF FARADAY'S LAW

SUMMARY : STATIONARY LOOP IN
TIME VARYING $\vec{B}$

$$\mathcal{V}_{\text{emf}} = - \iint_S \frac{\partial \vec{B}}{\partial t} \cdot d\vec{S} = \oint \vec{E} \cdot d\vec{\ell}$$

$$\vec{\nabla} \times \vec{E} = - \frac{\partial \vec{B}}{\partial t}$$

* ONE MAXWELL'S EQUATION FOR
TIME VARYING FIELDS

$$\Phi_B = \int_S \vec{B} \cdot \hat{n} da.$$

Example 6-1: Inductor in a Changing Magnetic Field

An inductor is formed by winding N turns of a thin conducting wire into a circular loop of radius a . The inductor loop is in the x - y plane with its center at the origin, and connected to a resistor R , as shown in Fig. 6-3. In the presence of a magnetic field $\mathbf{B} = B_0(\hat{\mathbf{y}}2 + \hat{\mathbf{z}}3) \sin \omega t$, where ω is the angular frequency, find

- (a) the magnetic flux linking a single turn of the inductor,
- (b) the transformer emf, given that $N = 10$, $B_0 = 0.2$ T, $a = 10$ cm, and $\omega = 10^3$ rad/s,
- (c) the polarity of $V_{\text{emf}}^{\text{tr}}$ at $t = 0$, and
- (d) the induced current in the circuit for $R = 1$ k Ω (assume the wire resistance to be much smaller than R).

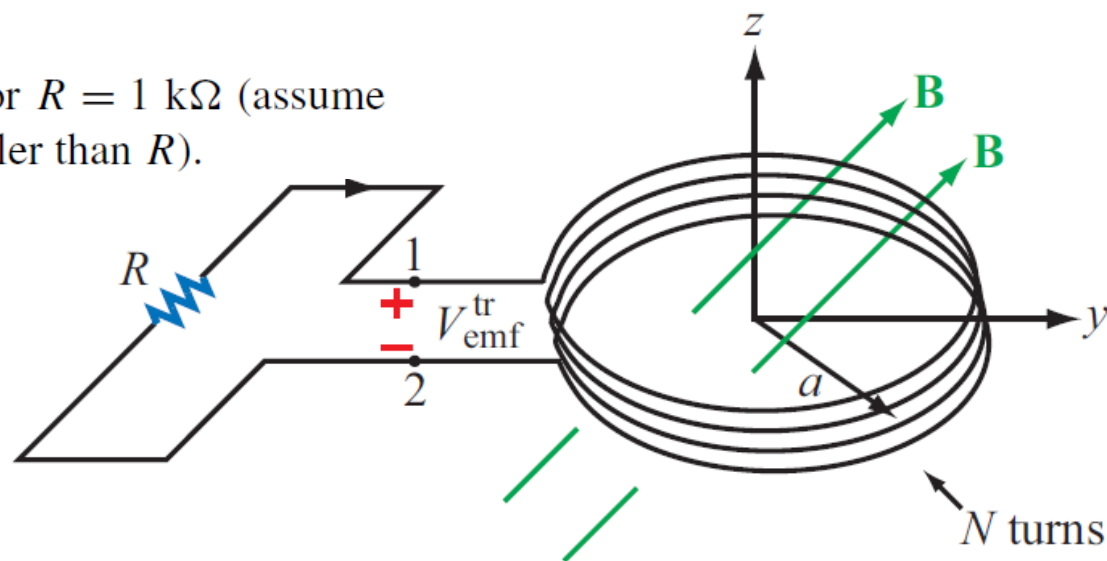


Figure 6-3: Circular loop with N turns in the x - y plane. The magnetic field is $\mathbf{B} = B_0(\hat{\mathbf{y}}2 + \hat{\mathbf{z}}3) \sin \omega t$ (Example 6-1).

Solution: (a) The magnetic flux linking each turn of the inductor is

$$\begin{aligned}\Phi &= \int_S \mathbf{B} \cdot d\mathbf{s} \\ &= \int_S [B_0(\hat{\mathbf{y}}2 + \hat{\mathbf{z}}3) \sin \omega t] \cdot \hat{\mathbf{z}} d\mathbf{s} \\ &= 3\pi a^2 B_0 \sin \omega t.\end{aligned}$$

(b) To find $V_{\text{emf}}^{\text{tr}}$, we can apply Eq. (6.8) or we can apply the general expression given by Eq. (6.6) directly. The latter approach gives

$$\begin{aligned}V_{\text{emf}}^{\text{tr}} &= -N \frac{d\Phi}{dt} \\ &= -\frac{d}{dt}(3\pi N a^2 B_0 \sin \omega t) \\ &= -3\pi N \omega a^2 B_0 \cos \omega t.\end{aligned}$$

For $N = 10$, $a = 0.1$ m, $\omega = 10^3$ rad/s, and $B_0 = 0.2$ T,

$$V_{\text{emf}}^{\text{tr}} = -188.5 \cos 10^3 t \quad (\text{V}).$$

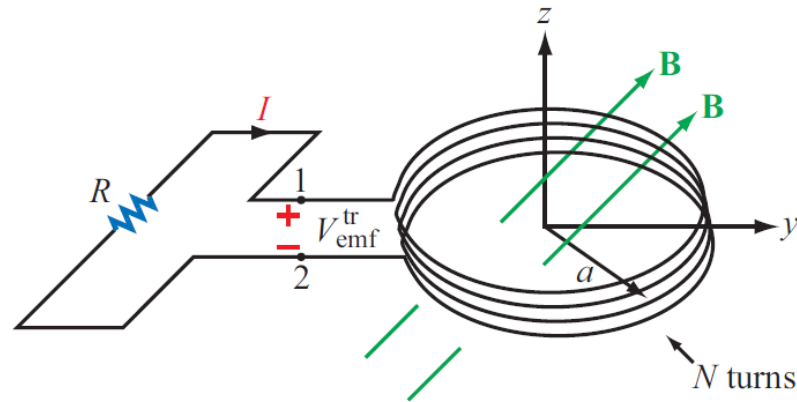


Figure 6-3: Circular loop with N turns in the x - y plane. The magnetic field is $\mathbf{B} = B_0(\hat{\mathbf{y}}2 + \hat{\mathbf{z}}3) \sin \omega t$ (Example 6-1).

(c) At $t = 0$, $d\Phi/dt > 0$ and $V_{\text{emf}}^{\text{tr}} = -188.5$ V. Since the flux is increasing, the current I must be in the direction shown in Fig. 6-3 in order to satisfy Lenz's law. Consequently, terminal 2 is at a higher potential than terminal 1 and

$$\begin{aligned}V_{\text{emf}}^{\text{tr}} &= V_1 - V_2 \\ &= -188.5 \quad (\text{V}).\end{aligned}$$

(d) The current I is given by

$$\begin{aligned}I &= \frac{V_2 - V_1}{R} \\ &= \frac{188.5}{10^3} \cos 10^3 t \\ &= 0.19 \cos 10^3 t \quad (\text{A}).\end{aligned}$$

Motional emf

MOVING CONDUCTOR IN A STATIC FIELD

Magnetic force on charge q moving with velocity $\mathbf{u}$ in a magnetic field $\mathbf{B}$:

$$\mathbf{F}_m = q(\mathbf{u} \times \mathbf{B}).$$

This magnetic force is equivalent to the electrical force that would be exerted on the particle by the electric field $\mathbf{E}_m$ given by

$$\mathbf{E}_m = \frac{\mathbf{F}_m}{q} = \mathbf{u} \times \mathbf{B}.$$

This, in turn, induces a voltage difference between ends 1 and 2, with end 2 being at the higher potential. The induced voltage is

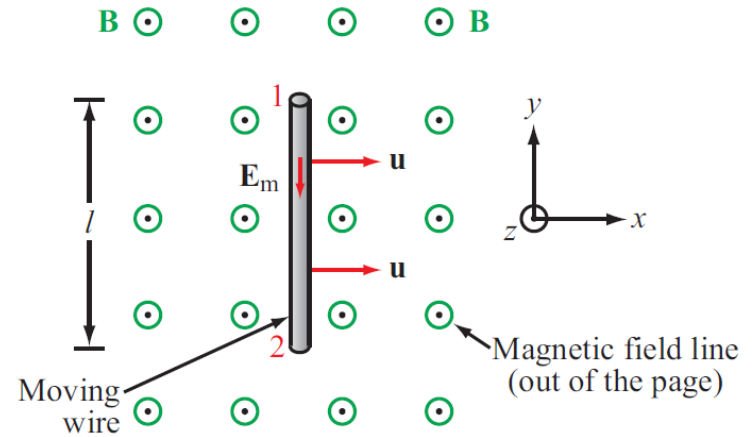


Figure 6-7: Conducting wire moving with velocity $\mathbf{u}$ in a static magnetic field.

For the conducting wire, $\mathbf{u} \times \mathbf{B} = \hat{x}u \times \hat{z}B_0 = -\hat{y}uB_0$ and $d\mathbf{l} = \hat{y} dl$. Hence,

$$V_{\text{emf}}^m = V_{12} = -uB_0l. \quad (6.25)$$

$$V_{\text{emf}}^m = V_{12} = \int_2^1 \mathbf{E}_m \cdot d\mathbf{l} = \int_2^1 (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l}.$$

2 IS AT A HIGHER POTENTIAL

Motional emf

In general, if any segment of a closed circuit with contour C moves with a velocity $\mathbf{u}$ across a static magnetic field $\mathbf{B}$, then the induced motional emf is given by

$$V_{\text{emf}}^{\text{m}} = \oint_C (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l} \quad (\text{motional emf}). \quad (6.26)$$

Only those segments of the circuit that cross magnetic field lines contribute to $V_{\text{emf}}^{\text{m}}$.

Example : Sliding Bar

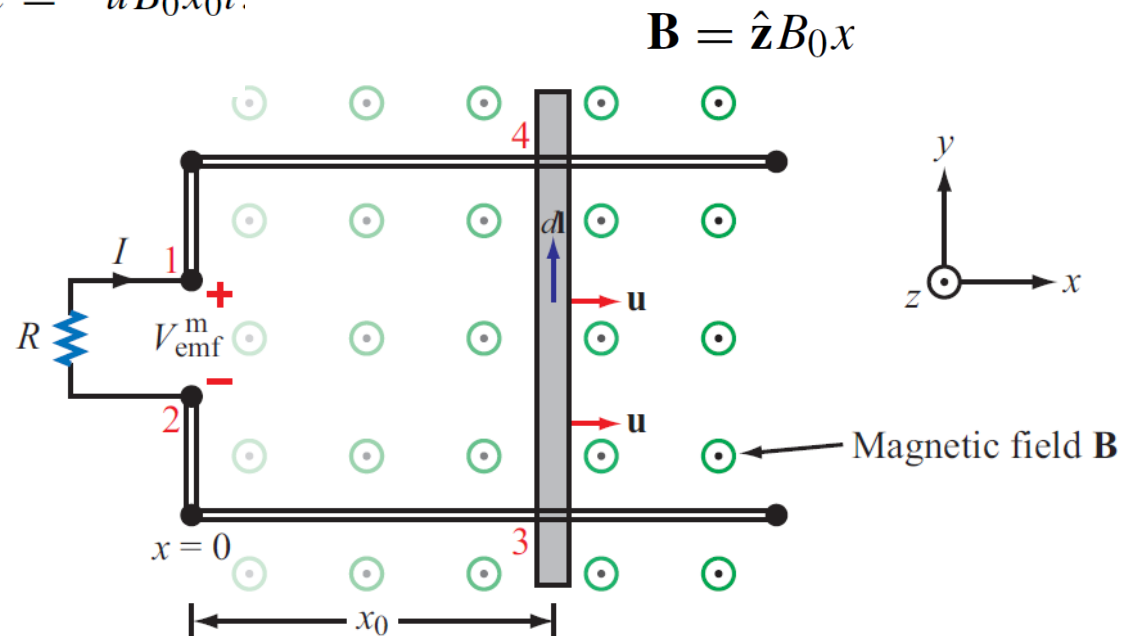
$$V_{\text{emf}}^{\text{m}} = V_{12} = V_{43} = \int_3^4 (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l}$$

$$= \int_3^4 (\hat{\mathbf{x}}u \times \hat{\mathbf{z}}B_0x_0) \cdot \hat{\mathbf{y}} dl = -uB_0x_0l.$$

Note that B increases with x

The length of the loop is related to u by $x_0 = ut$. Hence

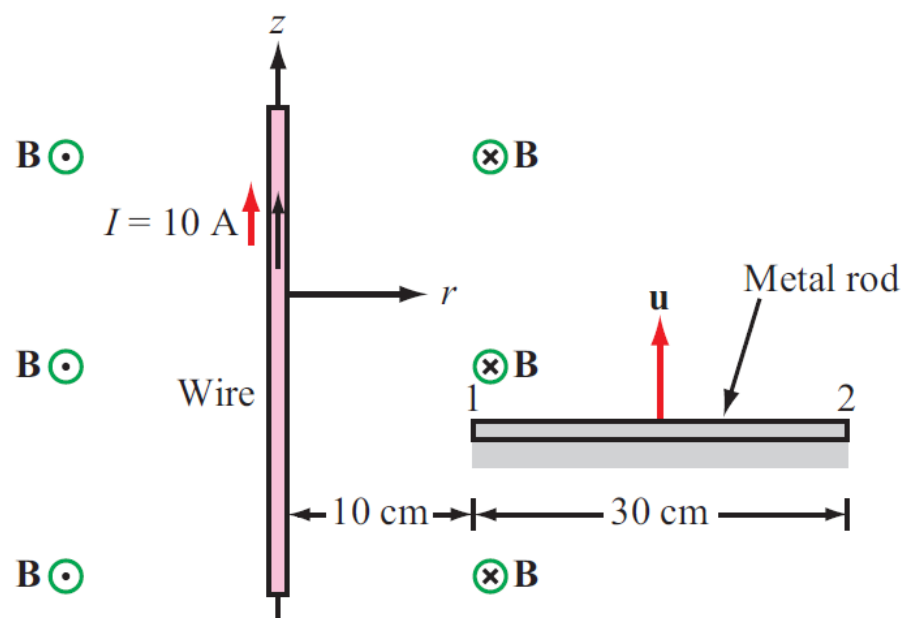
$$V_{\text{emf}}^{\text{m}} = -B_0u^2lt \quad (\text{V}).$$



Example 6-5: Moving Rod Next to a Wire

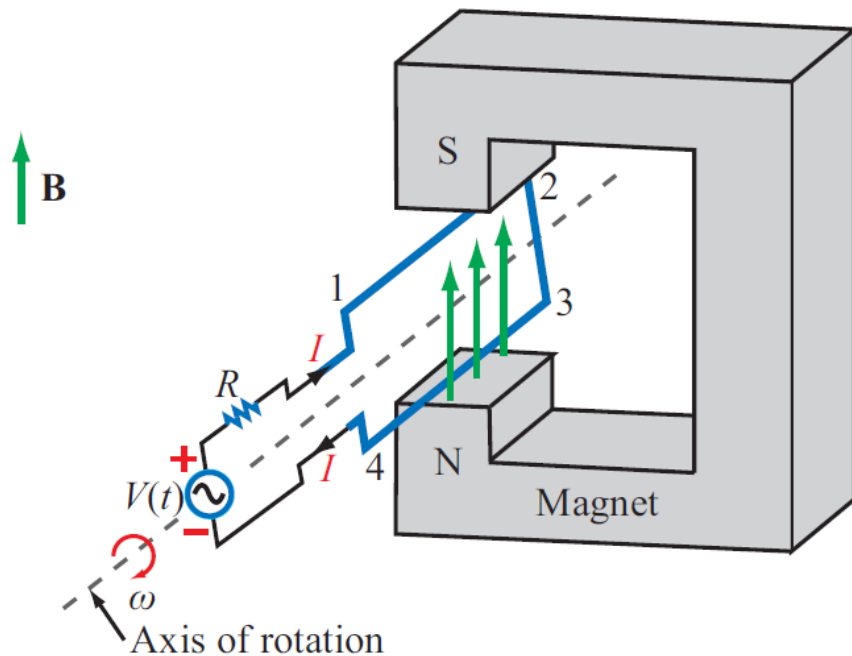
The wire shown in Fig. 6-10 carries a current $I = 10$ A. A 30-cm-long metal rod moves with a constant velocity $\mathbf{u} = \hat{\mathbf{z}}5$ m/s. Find V_{12} .

$$\mathbf{B} = \hat{\phi} \frac{\mu_0 I}{2\pi r}$$



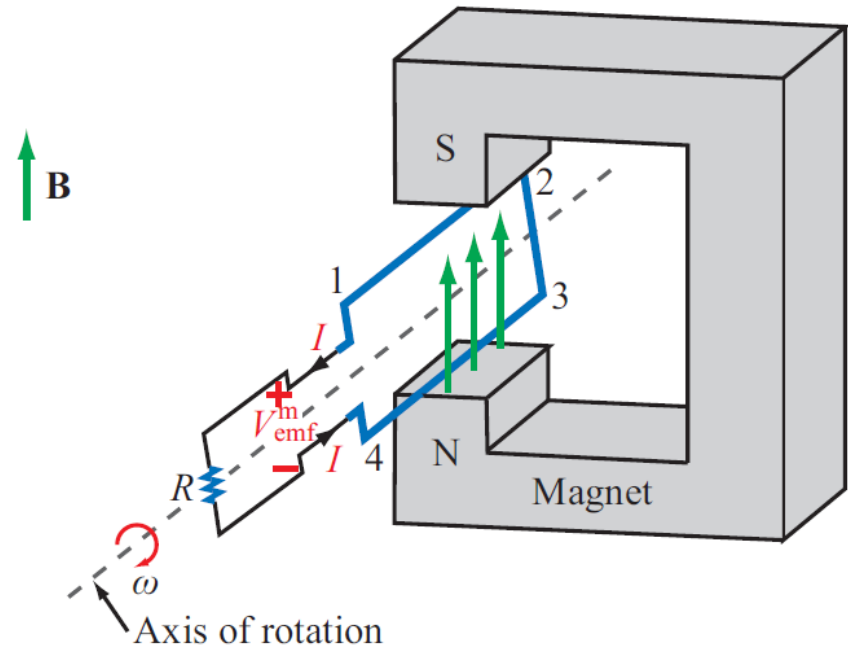
$$\begin{aligned} V_{12} &= \int_{40 \text{ cm}}^{10 \text{ cm}} (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l} \\ &= \int_{40 \text{ cm}}^{10 \text{ cm}} \left(\hat{\mathbf{z}}5 \times \hat{\phi} \frac{\mu_0 I}{2\pi r} \right) \cdot \hat{\mathbf{r}} dr \\ &= -\frac{5\mu_0 I}{2\pi} \int_{40 \text{ cm}}^{10 \text{ cm}} \frac{dr}{r} \\ &= -\frac{5 \times 4\pi \times 10^{-7} \times 10}{2\pi} \times \ln \left(\frac{10}{40} \right) \\ &= 13.9 \quad (\mu\text{V}). \end{aligned}$$

EM Motor/ Generator Reciprocity



(a) ac motor

Motor: Electrical to mechanical energy conversion



(b) ac generator

Generator: Mechanical to electrical energy conversion

Summary

Faraday's Law

$$V_{\text{emf}} = -\frac{d\Phi}{dt} = -\frac{d}{dt} \int_S \mathbf{B} \cdot d\mathbf{s} = V_{\text{emf}}^{\text{tr}} + V_{\text{emf}}^{\text{m}}$$

Transformer

$$V_{\text{emf}}^{\text{tr}} = -N \int_S \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s} \quad (N \text{ loops})$$

Motional

$$V_{\text{emf}}^{\text{m}} = \oint_C (\mathbf{u} \times \mathbf{B}) \cdot d\mathbf{l}$$